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Inventor(s)	Zhou; Shanyu et al.

Timing advance validation techniques for L1/L2 inter-cell mobility

Abstract

A method of wireless communication at a UE is disclosed herein. The method includes transmitting, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. The method includes receiving, at a second time instance after the first time instance, an estimated TA associated with the first cell. The method includes performing a second measurement on the at least one DL-RS. The method includes accepting or rejecting the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement.

Inventors: Zhou; Shanyu (San Diego, CA), Damnjanovic; Jelena (Del Mar, CA), Luo; Tao (San Diego, CA), Park; Changhwan (San Diego, CA)

Applicant: QUALCOMM Incorporated (San Diego, CA)

Family ID: 1000008819603

Assignee: QUALCOMM Incorporated (San Diego, CA)

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Primary Examiner: Zhao; Wei

Attorney, Agent or Firm: Procopio, Cory, Hargreaves & Savitch LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to communication systems, and more particularly, to timing advances (TAs).

INTRODUCTION

(2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division

multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

(3) These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

(4) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus for wireless communication at a user equipment (UE) are provided. The apparatus includes a memory and at least one processor coupled to the memory and based at least in part on information stored in the memory, the at least one processor is configured to: transmit, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, where the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; receive, at a second time instance after the first time instance, an estimated timing advance (TA) associated with the first cell; perform a second measurement on the at least one DL-RS; and accept or reject the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement.

(6) In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus for wireless communication at a network entity are provided. The apparatus includes a memory and at least one processor coupled to the memory and based at least in part on information stored in the memory, the at least one processor is configured to receive, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, where the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; estimate a timing advance (TA) associated with the first cell based on the first measurement on the at least one DL-RS; and transmit, at a second time instance after the first time instance, the estimated TA associated with the first cell.

(7) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating an example of a wireless communications system and an access network.
- (2) FIG. 2A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.
- (3) FIG. 2B is a diagram illustrating an example of downlink (DL) channels within a subframe, in accordance with various aspects of the present disclosure.
- (4) FIG. 2C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.
- (5) FIG. 2D is a diagram illustrating an example of uplink (UL) channels within a subframe, in accordance with various aspects of the present disclosure.
- (6) FIG. 3 is a diagram illustrating an example of a base station and user equipment (UE) in an access network.
- (7) FIG. 4 is a diagram illustrating example cell designations.
- (8) FIG. 5 is a diagram illustrating example timing adjustment groups (TAGs).
- (9) FIG. 6 is a diagram illustrating an example of a UE calculating a DL timing difference and receiving a TA.
- (10) FIG. 7 is a diagram illustrating an example of UE-side TA validation.
- (11) FIG. 8 is a diagram illustrating example communications between a UE and a base station.
- (12) FIG. 9 is a diagram illustrating example communications between a UE, a special cell (SpCell), and a to-be-activated cell.
- (13) FIG. 10 is a diagram illustrating example communications between a UE, a SpCell, and a to-be-activated cell.
- (14) FIG. 11 is a flowchart of a method of wireless communication.
- (15) FIG. 12 is a flowchart of a method of wireless communication.
- (16) FIG. 13 is a flowchart of a method of wireless communication.
- (17) FIG. 14 is a flowchart of a method of wireless communication.
- (18) FIG. 15 is a diagram illustrating an example of a hardware implementation for an example apparatus and/or network entity.
- (19) FIG. 16 is a diagram illustrating an example of a hardware implementation for an example network entity.

DETAILED DESCRIPTION

- (20) A UE may apply a TA for UL transmissions with a cell to account for propagation delays between the UE and the cell. When the UE receives signaling from a network to connect to a to-be-activated cell in a L1/L2 mobility configured cell set, the network may provide an estimated TA to the UE, where the estimated TA may be based on a measurement report transmitted by the UE to the network. However, the estimated TA may be inaccurate. Applying an inaccurate TA for UL transmissions may impact communications reliability. Various techniques for UE-side TA validation are described herein that may utilize reference signal received power (RSRP) measurements to represent channel strength. In an example, a UE transmits, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. The UE receives, at a second time instance after the first time instance, an estimated TA associated with the first cell. The UE performs a second measurement on the at least one DL-RS. The UE accepts or rejects the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. Via the aforementioned TA validation procedure, the UE may be able to reject inaccurate TAs. As such, the TA validation procedure may lead to increased communications reliability.
- (21) The detailed description set forth below in connection with the drawings describes various

configurations and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(22) Several aspects of telecommunication systems are presented with reference to various apparatus and methods. These apparatus and methods are described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(23) By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise, shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.

(24) Accordingly, in one or more example aspects, implementations, and/or use cases, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

(25) While aspects, implementations, and/or use cases are described in this application by illustration to some examples, additional or different aspects, implementations and/or use cases may come about in many different arrangements and scenarios. Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects, implementations, and/or use cases may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described examples may occur. Aspects, implementations, and/or use cases may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques herein. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described

aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.

(26) Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

(27) An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

(28) Base station operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

(29) FIG. 1 is a diagram **100** illustrating an example of a wireless communications system and an access network. The illustrated wireless communications system includes a disaggregated base station architecture. The disaggregated base station architecture may include one or more CUs **110** that can communicate directly with a core network **120** via a backhaul link, or indirectly with the core network **120** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **125** via an E2 link, or a Non-Real Time (Non-RT) RIC **115** associated with a Service Management and Orchestration (SMO) Framework **105**, or both). A CU **110** may communicate with one or more DUs **130** via respective midhaul links, such as an F1 interface. The DUs **130** may communicate with one or more RUs **140** via respective fronthaul links. The RUs **140** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **140**.

(30) Each of the units, i.e., the CUs **110**, the DUs **130**, the RUs **140**, as well as the Near-RT RICs **125**, the Non-RT RICs **115**, and the SMO Framework **105**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or to transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units,

or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or to transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), configured to receive or to transmit signals, or both, over a wireless transmission medium to one or more of the other units.

(31) In some aspects, the CU **110** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **110**. The CU **110** may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **110** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as an E1 interface when implemented in an O-RAN configuration. The CU **110** can be implemented to communicate with the DU **130**, as necessary, for network control and signaling.

(32) The DU **130** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **140**. In some aspects, the DU **130** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation, demodulation, or the like) depending, at least in part, on a functional split, such as those defined by 3GPP. In some aspects, the DU **130** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **130**, or with the control functions hosted by the CU **110**.

(33) Lower-layer functionality can be implemented by one or more RUs **140**. In some deployments, an RU **140**, controlled by a DU **130**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **140** can be implemented to handle over the air (OTA) communication with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **140** can be controlled by the corresponding DU **130**. In some scenarios, this configuration can enable the DU(s) **130** and the CU **110** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

(34) The SMO Framework **105** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **105** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements that may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **105** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **190**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **110**, DUs **130**, RUs **140** and Near-RT RICs **125**. In some implementations, the SMO Framework **105** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **111**, via an O1 interface. Additionally, in some implementations, the SMO Framework **105** can communicate directly with one or more RUs **140** via an O1 interface. The SMO Framework **105** also may include a Non-RT RIC **115** configured to

support functionality of the SMO Framework **105**.

(35) The Non-RT RIC **115** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, artificial intelligence (AI)/machine learning (ML) (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **125**. The Non-RT RIC **115** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **125**. The Near-RT RIC **125** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **110**, one or more DUs **130**, or both, as well as an O-eNB, with the Near-RT RIC **125**.

(36) In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **125**, the Non-RT RIC **115** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **125** and may be received at the SMO Framework **105** or the Non-RT RIC **115** from non-network data sources or from network functions. In some examples, the Non-RT RIC **115** or the Near-RT RIC **125** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **115** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **105** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

(37) At least one of the CU **110**, the DU **130**, and the RU **140** may be referred to as a base station **102**. Accordingly, a base station **102** may include one or more of the CU **110**, the DU **130**, and the RU **140** (each component indicated with dotted lines to signify that each component may or may not be included in the base station **102**). The base station **102** provides an access point to the core network **120** for a UE **104**. The base stations **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The small cells include femtocells, picocells, and microcells. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links between the RUs **140** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to an RU **140** and/or downlink (DL) (also referred to as forward link) transmissions from an RU **140** to a UE **104**. The communication links may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

(38) Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL wireless wide area network (WWAN) spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, Bluetooth, Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.

(39) The wireless communications system may further include a Wi-Fi AP **150** in communication

with UEs **104** (also referred to as Wi-Fi stations (STAs)) via communication link **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the UEs **104/AP 150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

(40) The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave” band.

(41) The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHz), FR4 (71 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

(42) With the above aspects in mind, unless specifically stated otherwise, the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.

(43) The base station **102** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate beamforming. The base station **102** may transmit a beamformed signal **182** to the UE **104** in one or more transmit directions. The UE **104** may receive the beamformed signal from the base station **102** in one or more receive directions. The UE **104** may also transmit a beamformed signal **184** to the base station **102** in one or more transmit directions. The base station **102** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **102/UE 104** may perform beam training to determine the best receive and transmit directions for each of the base station **102/UE 104**. The transmit and receive directions for the base station **102** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

(44) The base station **102** may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), network node, network entity, network equipment, or some other suitable terminology. The base station **102** can be implemented as an integrated access and backhaul (IAB) node, a relay node, a sidelink node, an aggregated (monolithic) base station with a baseband unit (BBU) (including a CU and a DU) and an RU, or as a disaggregated base station including one or more of a CU, a DU, and/or an RU. The set of base stations, which may include disaggregated base stations and/or aggregated base stations, may be referred to as next generation (NG) RAN (NG-RAN).

(45) The core network **120** may include an Access and Mobility Management Function (AMF) **161**, a Session Management Function (SMF) **162**, a User Plane Function (UPF) **163**, a Unified Data Management (UDM) **164**, one or more location servers **168**, and other functional entities. The

AMF **161** is the control node that processes the signaling between the UEs **104** and the core network **120**. The AMF **161** supports registration management, connection management, mobility management, and other functions. The SMF **162** supports session management and other functions. The UPF **163** supports packet routing, packet forwarding, and other functions. The UDM **164** supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The one or more location servers **168** are illustrated as including a Gateway Mobile Location Center (GMLC) **165** and a Location Management Function (LMF) **166**. However, generally, the one or more location servers **168** may include one or more location/positioning servers, which may include one or more of the GMLC **165**, the LMF **166**, a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like. The GMLC **165** and the LMF **166** support UE location services. The GMLC **165** provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The LMF **166** receives measurements and assistance information from the NG-RAN and the UE **104** via the AMF **161** to compute the position of the UE **104**. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UE **104**. Positioning the UE **104** may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UE **104** and/or the serving base station **102**. The signals measured may be based on one or more of a satellite positioning system (SPS) **170** (e.g., one or more of a Global Navigation Satellite System (GNSS), global position system (GPS), non-terrestrial network (NTN), or other satellite position/location system), LTE signals, wireless local area network (WLAN) signals, Bluetooth signals, a terrestrial beacon system (TBS), sensor-based information (e.g., barometric pressure sensor, motion sensor), NR enhanced cell ID (NR E-CID) methods, NR signals (e.g., multi-round trip time (Multi-RTT), DL angle-of-departure (DL-AoD), DL time difference of arrival (DL-TDOA), UL time difference of arrival (UL-TDOA), and UL angle-of-arrival (UL-AoA) positioning), and/or other systems/signals/sensors.

(46) Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.

(47) Referring again to FIG. **1**, in certain aspects, the UE **104** may include a TA validation component **198** that is configured to transmit, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set; receive, at a second time instance after the first time instance, an estimated TA associated with the first cell; perform a second measurement on the at least one DL-RS; and accept or reject the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. In certain aspects, the base station **102** may include a TA component **199** that is configured to receive, at a first time instance, an indication of a first measurement on at least one

DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set; estimate a TA associated with the first cell based on the first measurement on the at least one DL-RS; and transmit, at a second time instance after the first time instance, the estimated TA associated with the first cell. Although the following description may be focused on 5G NR, the concepts described herein may be applicable to other similar areas, such as LTE, LTE-A, CDMA, GSM, and other wireless technologies.

(48) FIG. 2A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. 2B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. 2C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. 2D is a diagram **280** illustrating an example of UL channels within a 5G NR subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. 2A, 2C, the 5G NR frame structure is assumed to be TDD, with subframe **4** being configured with slot format **28** (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe **3** being configured with slot format **1** (with all UL). While subframes **3**, **4** are shown with slot formats **1**, **28**, respectively, any particular subframe may be configured with any of the various available slot formats **0-61**. Slot formats **0**, **1** are all DL, UL, respectively. Other slot formats **2-61** include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.

(49) FIGS. 2A-2D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (also referred to as single carrier frequency-division multiple access (SC-FDMA) symbols) (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) (see Table 1). The symbol length/duration may scale with 1/SCS.

(50) TABLE-US-00001 TABLE 1 Numerology, SCS, and CP

SCS μ	$\Delta f = 2^{\mu} \cdot 15$ [kHz]	Cyclic prefix
0	15 Normal	1
1	30 Normal	2
2	60 Normal, Extended	3
3	120 Normal	4
4	240 Normal	5
5	480 Normal	6
6	960 Normal	

(51) For normal CP (14 symbols/slot), different numerologies μ 0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per subframe. Accordingly, for normal CP and numerology μ , there are 14 symbols/slot and 2^{μ} slots/subframe. The subcarrier spacing may be equal to $2^{\mu} \cdot 15$ kHz, where μ is the numerology 0 to 4. As such, the numerology $\mu=0$ has a subcarrier spacing of 15 kHz and the numerology $\mu=4$ has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. 2A-2D provide an example of normal CP with 14 symbols per slot and numerology $\mu=2$ with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μ s. Within a set of frames, there may be

one or more different bandwidth parts (BWPs) (see FIG. 2B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

(52) A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.

(53) As illustrated in FIG. 2A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).

(54) FIG. 2B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE **104** to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and paging messages.

(55) As illustrated in FIG. 2C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.

(56) FIG. 2D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.

(57) FIG. 3 is a block diagram of a base station **310** in communication with a UE **350** in an access

network. In the DL, Internet protocol (IP) packets may be provided to a controller/processor **375**. The controller/processor **375** implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor **375** provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

(58) The transmit (TX) processor **316** and the receive (RX) processor **370** implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor **316** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator **374** may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **350**. Each spatial stream may then be provided to a different antenna **320** via a separate transmitter **318Tx**. Each transmitter **318Tx** may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.

(59) At the UE **350**, each receiver **354Rx** receives a signal through its respective antenna **352**. Each receiver **354Rx** recovers information modulated onto an RF carrier and provides the information to the receive (RX) processor **356**. The TX processor **368** and the RX processor **356** implement layer 1 functionality associated with various signal processing functions. The RX processor **356** may perform spatial processing on the information to recover any spatial streams destined for the UE **350**. If multiple spatial streams are destined for the UE **350**, they may be combined by the RX processor **356** into a single OFDM symbol stream. The RX processor **356** then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **310**. These soft decisions may be based on channel estimates computed by the channel estimator **358**. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station **310** on the physical channel. The data and control signals are then provided to the controller/processor **359**, which implements

layer 3 and layer 2 functionality.

(60) The controller/processor **359** can be associated with a memory **360** that stores program codes and data. The memory **360** may be referred to as a computer-readable medium. In the UL, the controller/processor **359** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets. The controller/processor **359** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

(61) Similar to the functionality described in connection with the DL transmission by the base station **310**, the controller/processor **359** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

(62) Channel estimates derived by a channel estimator **358** from a reference signal or feedback transmitted by the base station **310** may be used by the TX processor **368** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor **368** may be provided to different antenna **352** via separate transmitters **354Tx**. Each transmitter **354Tx** may modulate an RF carrier with a respective spatial stream for transmission.

(63) The UL transmission is processed at the base station **310** in a manner similar to that described in connection with the receiver function at the UE **350**. Each receiver **318Rx** receives a signal through its respective antenna **320**. Each receiver **318Rx** recovers information modulated onto an RF carrier and provides the information to a RX processor **370**.

(64) The controller/processor **375** can be associated with a memory **376** that stores program codes and data. The memory **376** may be referred to as a computer-readable medium. In the UL, the controller/processor **375** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets. The controller/processor **375** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

(65) At least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359** may be configured to perform aspects in connection with the TA validation component **198** of FIG. **1**.

(66) At least one of the TX processor **316**, the RX processor **370**, and the controller/processor **375** may be configured to perform aspects in connection with the TA component **199** of FIG. **1**.

(67) FIG. **4** is a diagram **400** that illustrates an example of dual connectivity and carrier aggregation (CA). A UE **402** may be connected to a master cell group (MCG) **404** and a secondary cell group (SCG) **406**. Such an arrangement may be referred to as dual connectivity (DC) **408**. The MCG **404** may be a group of serving cells associated with a master node that has a control plane connection to a core network (e.g., the core network **120**). The master node may be a base station such as a gNB, an eNB, etc. (i.e., a network entity). The SCG **406** may be a group of serving cells associated with a secondary node that does not have a control plane connection to the core network. The secondary node may be a base station, such as a gNB, an eNB, etc. (i.e., a network entity). The MCG **404** may become the SCG **406** and the SCG **406** may become the MCG **404** based upon various factors, such as a location of the UE **402**, network conditions, etc.

(68) The MCG **404** includes a PCell **410**. The MCG **404** may also include one or more SCells (e.g.,

a first SCell **412** and an Nth SCell **414**, where N is a positive integer greater than one). The PCell **410**, the first SCell **412**, and/or the Nth SCell **414** may be in a CA **416** configuration. In some aspects, a primary cell may become a secondary cell or that a secondary cell may become a primary cell based upon various factors, such as a location of the UE **402**, network conditions, etc.

(69) The SCG **406** includes a primary secondary cell (PSCell) **418**. The PSCell **418** is a primary cell of the SCG **406**. The SCG **406** may also include one or more SCells (e.g., a first SCell **420** and an Mth SCell **422**, where M is a positive integer greater than one). The PSCell **418**, the first SCell **420**, and/or the Mth SCell **422** may be in a CA configuration **424**. In some aspects, a primary secondary cell may become a secondary cell or that a secondary cell may become a primary secondary cell based upon various factors, such as a location of the UE **402**, network conditions, etc.

(70) The PCell **410** of the MCG **404** and the PSCell **418** of the SCG **406** may be referred to as a special cell (SpCell) **426**. For a dual connectivity operation, the term SpCell refers to the PCell of the MCG or the PSCell of the SCG, e.g., depending on whether the MAC entity is associated to the MCG or the SCG, respectively. An SpCell supports PUCCH transmission and contention-based random access, and is always activated. The SpCell **426** is responsible for data and control functionality. In cases in which there is no DC (e.g., when the MCG **404** is configured and when the SCG **406** is not configured), the PCell **410** may be referred to as the SpCell **426**. A cell group that includes the SpCell **426** may be referred to as a PCG. It is to be understood that an SpCell may change based upon various factors, such as a location of the UE **402**, network conditions, etc. In an example, a network entity may configure the first SCell **412** as a primary cell and Mth SCell **422** as a primary secondary cell to configure a new SpCell.

(71) FIG. 5 is a diagram **500** illustrating example TAGs for cell synchronization. A base station **504** (e.g., via a DU **506**) may configure a L1/L2 mobility configured cell set **508** for a UE **502**. The term “L1/L2 mobility configured cell set” may refer to a set of cells configured for a UE such that cells in the set of cells may be updated to become an SpCell by a base station (e.g., a gNB) using L1 or L2 signaling. L1 signaling may refer to DCI. L2 signaling may refer to a medium access control (MAC) control element (MAC-CE). Cells (represented as circles in the diagram **500**) within the L1/L2 mobility configured cell set **508** may belong to the same or different TAGs. Each TAG may be associated with a different TA. The term “TA” may refer to a command sent by a base station to a UE to adjust an uplink transmission of the UE such that the UE sends UL symbols in advance according to a command for a transmission, such as a PUSCH, PUCCH, and/or SRS transmission. The UE **502** may apply a TA associated with a TAG for uplink communications via/with/over a cell belonging to the TAG. A TA may refer to a negative offset at the UE **502** between a start of a DL slot as observed by the UE **502** and a start of an UL slot. The TA may account for propagation delays such that UL and DL transmission of the UE align at a reference point. In general, a first UE that is located relatively far from a base station may start an UL transmission relatively sooner than a second UE that is located relatively near the base station. As such, the first UE may apply a larger TA for UL transmissions than a TA applied by the second UE.

(72) A TAG that includes a PCell (or an SpCell) may be referred to as a primary TAG (pTAG). A TAG that includes SCell(s) (and not a PCell or an SpCell) may be referred to as a secondary TAG (sTAG). The diagram **500** depicts cells belonging to a pTAG, cells belonging to a sTAG1, and cells belonging to a sTAG2. The base station **504** may associate SCells to a TAG when the SCells are added to the L1/L2 mobility configured cell set **508**. Initial TAG assignment may be determined by the base station **504** (i.e., a serving gNB). The base station **504** may determine the initial TAG assignment based upon a band of operations, existence of repeaters, a location of a cell, and/or a location of the UE **502**.

(73) To establish initial timing, the UE **502** may perform a RACH procedure with the pTAG, that is, the UE **502** may perform the RACH procedure with the SpCell (or the PCell). The UE may obtain a TA associated with the pTAG upon completion of the RACH procedure. To further

establish initial timing, the UE **502** may perform a RACH procedure with a SCell belonging to a sTAG. For instance, the UE **502** may receive a PDCCH order from a base station associated with the SCell. The UE **502** may perform a contention-free RACH procedure upon reception of the PDCCH order. The UE **502** may obtain a TA associated with the sTAG upon completion of the contention-free RACH procedure.

(74) Cells (or cell groups) within the L1/L2 mobility configured cell set **508** may be activated or deactivated by the base station **504** via L1/L2 signaling. Cells/cell groups within the L1/L2 mobility configured cell set **508** that are activated may be used by the UE **502** for data transfer and control. In an example, when the UE **502** is at a first location, cells in the pTAG and the sTAG1 may be activated. As the UE **502** moves to a second location, the base station **504** may transmit L1/L2 signaling that activates/deactivates cells/cell groups. In an example, the L1/L2 signaling transmitted by the base station **504** when the UE **502** is at the second location may activate cells in the sTAG2.

(75) For L1/L2 inter-cell mobility, the UE **502** may obtain a TA for a cell/cell group when the cell/cell group is activated. Obtaining TAs for activated cells in a relatively rapid manner may facilitate increased communications reliability at the UE **502**. The UE **502** may receive frequent updates for TAs for activated cells/cell groups. However, TAs for deactivated cells/cell groups may not be as up-to-date as compared to TAs for activated cells/cell groups. In an example, the UE **502** may transmit a report (e.g., a L1 report) to the base station **504** that includes a DL timing difference between a reference cell and a deactivated cell. The base station **504** may calculate an estimated TA for the deactivated cell based upon the DL timing difference. The estimated TA for the deactivated cell may be inaccurate due to a lack of active communication between the UE **502** and the deactivated cell. The estimated TA for the deactivated cell may also be inaccurate due to a TA estimate error based on the DL timing difference included in the report. The UE **502** may obtain an accurate TA for a deactivated cell by performing a RACH procedure ordered by a PDCCH order; however, the RACH procedure may be time consuming.

(76) FIG. **6** is a diagram **600** illustrating an example of a UE **602** calculating a DL timing difference and receiving a TA. The UE **602** may perform validation on a TA obtained from a base station (e.g., a gNB) before applying the TA. The validation may be based on a quality of service (QoS) of serving traffic. The validation may also be based on a maximum transmission timing deviation allowance of the UE **602** (which may be referred to as “Te”). In an example, the maximum transmission timing deviation allowance may be 3 μ s.

(77) The UE **602** may perform reference signal received power (RSRP) measurements on downlink reference signals (DL-RSs) transmitted by a first deactivated cell **606**, an SpCell **604** for the UE **602**, and a second deactivated cell **608**. The UE **602** may also calculate DL timing differences (with respect to a reference cell) for the first deactivated cell **606**, the SpCell **604**, and the second deactivated cell **608** based on the DL-RSs.

(78) In an example, the UE **602** may receive a DL-RS from the first deactivated cell **606** and at time t1, the UE **602** may measure a DL timing difference with respect to a reference cell and report the DL timing difference to the SpCell **604**. At time t2 that occurs after time t1, the UE **602** may receive an estimated TA (referred to as “TA info” in the diagram **600**) from the SpCell **604**, where the estimated TA is for communications via/with/over the first deactivated cell **606**. In an example, the first deactivated cell **606** may be a cell that is to be activated. However, the UE **602** may have changed locations between time t1 and time t2. The estimated TA may be based on the DL timing difference reported at time t1. If a difference between time t2 and time t1 is greater than a threshold, the estimated TA may be inaccurate for communications via/with/over the first deactivated cell **606**. Stated differently, a strength of a channel associated with the first deactivated cell **606** may have changed by more than a threshold amount between time t1 and time t2. Furthermore, the UE **602** may not report (i.e., RSRP report) at a frequency that enables the SpCell **604** to accurately estimate the TA. Additionally, if the UE **602** does not apply the estimated TA

within a threshold period of time after receiving the estimated TA, the estimated TA may be inaccurate due to mobility of the UE **602**.

(79) In a wireless communication system, a network may aim to ensure that a UE maintains connectivity with a network entity (e.g., a base station) as the UE moves within a network. L1/L2 based inter-cell mobility may help to reduce mobility latency. For instance, configuration and maintenance of multiple candidate cells may allow for rapid application of configurations for candidate cells. Furthermore, dynamic switching mechanisms among candidate serving cells (including an SpCell and SCell) based on L1/L2 signaling may further reduce latency.

(80) L1/L2 based inter-cell mobility may be applicable to standalone operation scenarios, CA scenarios, and DC (e.g., NR-DC) scenarios. In the standalone operation scenarios and CA scenarios, L1/L2 based inter-cell mobility may be applicable for intra-DU cases and/or intra CU inter-DU cases. L1/L2 based inter-cell mobility is applicable in both FR1 and FR2, as well as both in intra-frequency and inter-frequency. In L1/L2 based inter-cell mobility, source cells and target cells may be synchronized or non-synchronized.

(81) A UE may apply a TA for UL transmissions with a cell to account for propagation delays between the UE and the cell. When the UE receives signaling from a network to connect to a to-be-activated cell in a L1/L2 mobility configured cell set, the network may provide an estimated TA to the UE, where the estimated TA may be based on a measurement report transmitted by the UE to the network. However, the estimated TA may be inaccurate for various reasons. In one example, channel strength may have changed between a time at which the UE last transmitted a measurement report and a time at which the estimated TA is obtained, which may lead to the estimated TA being inaccurate. In another example, the UE may not transmit measurement reports frequently enough to enable the network to accurately generate the estimated TA. In yet another example, the estimated TA may be inaccurate due to UE mobility, that is, a time at which the UE applies the estimated TA may be too far away from a time at which the network generates the estimated TA. Applying an inaccurate TA for UL transmissions may impact communications reliability.

(82) Various techniques for UE-side TA validation are described herein that may utilize RSRP measurements to represent channel strength. In an example, a UE transmits, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. The UE receives, at a second time instance after the first time instance, an estimated TA associated with the first cell. The UE performs a second measurement on the at least one DL-RS. The UE accepts or rejects the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. Via the aforementioned TA validation procedure, the UE may be able to reject inaccurate TAs. As such, the TA validation procedure may lead to increased communications reliability.

(83) FIG. 7 is a diagram **700** illustrating an example of UE-side TA validation. The UE-side TA validation may be based on RSRP measurements performed on DL-RSs. A UE **702** may utilize a filtered (e.g., averaged) RSRP measurement to represent channel strength. The channel strength may be associated with a location of the UE **702**. For instance, as the location of the UE **702** changes, a TA of the UE **702** for communication via/with/over a cell may change. Furthermore, as the location of the UE **702** changes, a RSRP measurement performed on a DL-RS associated with the cell may also change. In general, the UE **702** may utilize a change in RSRP measurement to determine a TA change.

(84) The UE **702** may be configured via RRC signaling with a RSRP change threshold. Each cell in a L1/L2 mobility configured cell set (e.g., the L1/L2 mobility configured cell set **508**) may be configured with a unique threshold (or the same threshold). At time t1, the UE **702** may transmit a L1 report to an SpCell **704**. The L1 report may include a downlink timing difference calculated by the UE **702** with respect to a reference cell. The UE **702** may calculate the DL timing difference

based on a DL-RS transmitted by a deactivated cell **706** in the L1/L2 mobility configured cell set (e.g., the L1/L2 mobility configured cell set **508**). The L1 report may also include a first RSRP measurement performed by the UE **702** on the DL-RS of the deactivated cell **706**.

(85) At time t_2 that occurs after time t_1 , the UE **702** may receive an estimated TA from the SpCell **704**, where the estimated TA may be based on the DL timing difference. In an example, the UE **702** has changed locations between time t_1 and time t_2 . As a result, the deactivated cell **706** may be activated at a time occurring after time t_2 . The estimated TA may be for communication of the UE **702** via/with/over the deactivated cell **706** (which is to be activated). The SpCell **704** may estimate the estimated TA based on the DL timing difference. Upon receiving the estimated TA, the UE **702** may perform a second RSRP measurement on the DL-RS transmitted by the deactivated cell **706**. The UE **702** may determine an RSRP change between time t_1 and time t_2 . For instance, the UE **702** may compare an absolute value of a difference between the first RSRP measurement at time t_1 and the second RSRP measurement at time t_2 with a threshold. The threshold may be unique to the deactivated cell **706** or shared by several cells. If the RSRP change is less than the threshold, the UE **702** may accept the estimated TA for communications via/with/over the (to-be-activated) deactivated cell **706**. If the RSRP change is greater than or equal to the threshold, the UE **702** may reject the estimated TA.

(86) FIG. **8** is a diagram **800** illustrating example communications between a UE **802** and a base station **804**. The communications may be for validating a TA. The base station **804** may be associated with a current SpCell of the UE **802**. At **806**, the UE **802** may receive a L1/L2 mobility configuration transmitted by the base station **804**. In an example, the L1/L2 mobility configuration may configure the L1/L2 mobility configured cell set **508**. At **808**, the UE **802** may receive, via RRC signaling, RSRP change thresholds for cells. The RRC signaling may configure each cell in the cells with an individual change threshold. At **809**, the UE **802** may transmit an acknowledgment to the base station **804**, where the acknowledgment may indicate that RRC configuration/reconfiguration is complete.

(87) At **810**, the UE **802** may perform, at a first point in time, first measurement(s) on a DL-RS associated with a cell. In an example, the cell may be a currently deactivated cell in the L1/L2 mobility configured cell set **508**. The first measurement(s) may include a DL timing difference of the cell with respect to a reference cell. The first measurement(s) may also include an RSRP measurement performed on the DL-RS. The RSRP measurement may be an average RSRP measurement.

(88) At **812**, the UE **802** may transmit the first measurement(s) to the base station **804**. At **814**, the base station **804** may estimate a TA for the cell based on the DL timing difference transmitted at **812**. At **816**, the UE **802** may receive the estimated TA from the base station **804**. At **818**, the UE **802** may perform, at a second point in time occurring after the first point in time, second measurement(s) on the DL-RS associated with the cell. The second measurement(s) may include an RSRP measurement. The RSRP measurement may be an average RSRP measurement. At **820**, the UE **802** may accept or reject the estimated TA for communication via/with/over the cell based on at least one of the first measurement(s) or the second measurement(s). In an example, the UE **802** may compute a difference between a first RSRP measurement performed by the UE **802** at **810** and a second RSRP measurement performed by the UE **802** at **818**. If the difference is greater than a threshold value, the UE **802** may reject the estimated TA. If the difference is less than the threshold value, the UE **802** may accept the estimated TA. In one aspect, the UE may compute an absolute value of the difference prior to comparing the difference to the threshold value. As noted above, the threshold value may be specific to the cell and may be configured via RRC signaling. If the UE **802** accepts the estimated TA, the UE **802** may communicate via/with/over the cell using the estimated TA.

(89) FIG. **9** is a diagram **900** illustrating example communications between a UE **902**, a SpCell **904**, and a to-be-activated cell **906**. The to-be-activated cell **906** may serve as an SpCell when

activated or as an SCell when activated. At **908**, the UE **902** may receive a new cell activation command for the to-be-activated cell **906** from the SpCell **904**. In an example, the SpCell **904** may be associated with the base station **804** described above in the description of the diagram **800**. The new cell activation command may include an estimated TA for the to-be-activated cell **906**. In an example, **908** may correspond to **816** described above in the description of the diagram **800**. The new cell activation command may indicate that the UE **902** is to acknowledge the new cell activation command. At **910**, the UE **902** may perform TA validation based upon the estimated TA included in the new cell activation command. In an example, **910** may correspond to **820** (and also **810** and **818**) described above in the description of the diagram **800**.

(90) In an example, the UE **902** may determine that the estimated TA is invalid (e.g., based on an RSRP change measurement) for communication via/with/over the to-be-activated cell **906**. At **912**, the UE **902** may transmit a TA invalid notification to the SpCell **904**. The TA invalid notification may indicate that the estimated TA has been rejected for communication with the to-be-activated cell **906**. In one aspect, the TA invalid notification is transmitted via a medium access control (MAC) control element (MAC-CE), such as a dedicated MAC-CE. In another aspect, the TA invalid notification is transmitted via an uplink control information (UCI), such as a dedicated UCI. In yet another aspect, the UE **902** may bind the TA-invalid notification with an acknowledgement/non-acknowledgment (ACK/NACK) message for the new cell activation command received at **908**. For instance, the UE **902** may delay acknowledging the new cell activation command until after TA validation is performed at **910**.

(91) In one aspect, at **914A**, the UE **902** may receive a PDCCH order from the SpCell **904** that orders the UE **902** to initiate a RACH procedure with the to-be-activated cell **906**. The SpCell **904** may transmit the PDCCH order based upon receiving the TA invalid notification transmitted by the UE **902** at **912**. In another aspect, at **914B**, the UE **902** may receive a PDCCH order from the to-be-activated cell **906** that orders the UE **902** to initiate a RACH procedure with the to-be-activated cell **906**. In an example, a base station associated with the SpCell **904** (e.g., the base station **804**) may communicate with a base station associated with the to-be-activated cell **906** in order to cause the to-be-activated cell **906** to transmit the PDCCH order received by the UE **902** at **914B**. At **916**, the UE **902** may initiate a RACH procedure with the to-be-activated cell **906** if the to-be-activated cell **906** is to serve as the new SpCell. The UE **902** may obtain a correct TA for the to-be-activated cell **906** via the RACH procedure. When the to-be-activated cell **906** is activated, the UE **902** may communicate via/with/over the (now-activated) to-be-activated cell **906** via the correct TA.

(92) FIG. **10** is a diagram **1000** illustrating example communications between a UE **1002**, a SpCell **1004**, and a to-be-activated cell **1006**. The to-be-activated cell **1006** may serve an SpCell when activated or as an SCell when activated. At **1008**, the UE **1002** may receive a new cell activation command for the to-be-activated cell **1006** from the SpCell **1004**. In an example, the SpCell **1004** may be associated with the base station **804** described above in the description of the diagram **800**. The new cell activation command may include an estimated TA for the to-be-activated cell **1006**. In an example, **1008** may correspond to **816** described above in the description of the diagram **800**. The new cell activation command may indicate that the UE **1002** is to acknowledge the new cell activation command. At **1010**, the UE **902** may perform TA validation based upon the estimated TA included in the new cell activation command. In an example, **1010** may correspond to **820** (and also **810** and **818**) described above in the description of the diagram **800**. For instance, the UE **1002** may perform first measurement(s) on a DL-RS associated with the to-be-activated cell **1006**, where the first measurement(s) may include a DL timing difference of the to-be-activated cell **1006** with respect to a reference cell and/or a RSRP measurement. The UE **1002** may transmit the first measurement(s) to the SpCell **1004** and receive an estimated TA from the SpCell **1004** based upon the DL timing difference.

(93) In an example, the UE **1002** may determine that the estimated TA is invalid (e.g., based on an RSRP change measurement) for communication via/with/over the to-be-activated cell **1006**. At

1011, the UE **1002** may perform another measurement on the DL-RS associated with the SpCell **1004**. Such a measurement may include a DL timing difference with respect to a reference cell and/or a RSRP measurement. At **1012**, the UE **902** may transmit a TA invalid notification. The TA invalid notification may indicate that the estimated TA has been rejected for communication with the to-be-activated cell **1006**. The TA invalid notification may also include UL synchronization assistance information (referred to herein as “UL sync assistance information”). The UL sync assistance information may include a latest DL timing difference (i.e., the DL timing difference measured at **1011**) for the to-be-activated cell **1006** with respect to a reference cell. In one aspect, the TA invalid notification is transmitted via MAC-CE, such as dedicated MAC-CE. In another aspect, the TA invalid notification is transmitted via a UCI, such as a dedicated UCI. In yet another aspect, the UE **1002** may bind the TA-invalid notification with ACK/NACK message for the new cell activation command received at **1008**. For instance, the UE **1002** may delay acknowledging the new cell activation command until after TA validation is performed at **1010**.

(94) At **1014**, the SpCell **1004** may estimate a new TA for communication via/with/over the to-be-activated cell **1006** based on the UL sync assistance information. At **1016**, the UE **1002** may receive the new estimated TA from the SpCell **1004**. In one aspect, the UE **1002** may receive the new estimated TA via a MAC-CE. In another aspect, the UE **1002** may receive the new estimated TA via a DCI. The UE **1002** may also receive an indication of RACH resources for use in a RACH procedure if the UE **1002** rejects the new estimated TA.

(95) At **1018**, the UE **1002** may perform TA validation based upon the new estimated TA. In an example, **1018** may correspond to **1010** (and **818/820**), with the new estimated TA being utilized in place of the (previous) estimated TA. For instance, the UE **1002** may perform a RSRP measurement on the DL-RS associated with the SpCell **1004** when the UE **1002** receives the new estimated TA. The UE **1002** may accept or reject the estimated TA based upon the RSRP measurement, the RSRP measurement performed at **1011**, and a threshold similar to the UE **802** at **820**. In an example, the UE **1002** may determine that the new estimated TA is invalid (e.g., based on an RSRP change measurement) for communication via/with/over the to-be-activated cell **1006**. At **1020**, the UE **1002** may determine whether a maximum number of TA rejections has occurred. In an example, the UE **1002** may receive an indication of the maximum number of TA rejections during the RRC configuration occurring at **808** in the diagram **800**. If the maximum number of TA rejections has not occurred, the UE **1002** may repeat **1012** with updated UL sync assistance information. If the maximum number of TA rejection has occurred, at **1024**, the UE **1002** may initiate a RACH procedure with the to-be-activated cell **1006** if the to-be-activated cell **1006** is to serve as a new SpCell. The UE **1002** may utilize the RACH resources received at **1016** for the RACH procedure. The UE **1002** may obtain a correct TA for the to-be-activated cell **1006** via the RACH procedure. When the to-be-activated cell **1006** is activated, the UE **902** may communicate via/with/over the (now-activated) to-be-activated cell **1006** via the correct TA.

(96) FIG. **11** is a flowchart **1100** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, the UE **350**, the UE **702**, the UE **802**, the UE **902**, the UE **1002**, the apparatus **1504**, etc.). The method may be performed by the TA validation component **198**. The method may be associated with various technical advantages at the UE, such as improved communications reliability via an accurate TA.

(97) At **1102**, the UE transmits, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. For example, FIG. **8** at **812** shows the UE **802** transmitting first measurement(s). In an example, the first measurement(s) may be performed on a DL-RS associated with the deactivated cell **706** in FIG. **7**. In another example, the first measurement(s) may include a DL timing difference with respect to a reference cell at time t_1 as illustrated in FIG. **6**. In a further example, the first measurement may also include a RSRP measurement performed at time t_1 as illustrated in FIG. **7**. In yet another example, the first cell may

be included in the L1/L2 mobility configured cell set **508** illustrated in FIG. 5. For example, **1102** may be performed by the TA validation component **198**.

(98) At **1104**, the UE receives, at a second time instance after the first time instance, an estimated TA associated with the first cell. For example, FIG. 8 at **816** shows the UE **802** receiving an estimated TA. In an example, the first cell may be the deactivated cell **706** in FIG. 7. For example, **1104** may be performed by the TA validation component **198**.

(99) At **1106**, the UE performs a second measurement on the at least one DL-RS. For example, FIG. 8 at **818** shows the UE **802** performing second measurement(s) on a DL-RS. In another example, the second measurement may include a RSRP measurement performed at time t2 as illustrated in FIG. 7. For example, **1106** may be performed by the TA validation component **198**.

(100) At **1108**, the UE accepts or rejects the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. For example, FIG. 8 at **820** shows the UE **802** accepting or rejecting an estimated TA for communication with a cell based on at least one of first measurement(s) or second measurement(s). For example, **1108** may be performed by the TA validation component **198**.

(101) FIG. 12 is a flowchart **1200** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, the UE **350**, the UE **702**, the UE **802**, the UE **902**, the UE **1002**, the apparatus **1504**, etc.). The method (including the various aspects described below) may be performed by the TA validation component **198**. The method may be associated with various technical advantages at the UE, such as improved communications reliability via an accurate TA.

(102) At **1204**, the UE transmits, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. For example, FIG. 8 at **812** shows the UE **802** transmitting first measurement(s). In an example, the first measurement(s) may be performed on a DL-RS associated with the deactivated cell **706** in FIG. 7. In another example, the first measurement(s) may include a DL timing difference with respect to a reference cell at time t1 as illustrated in FIG. 6. In a further example, the first measurement may also include a RSRP measurement performed at time t1 as illustrated in FIG. 7. In yet another example, the first cell may be included in the L1/L2 mobility configured cell set **508** illustrated in FIG. 5. For example, **1204** may be performed by the TA validation component **198**.

(103) At **1206**, the UE receives, at a second time instance after the first time instance, an estimated TA associated with the first cell. For example, FIG. 8 at **816** shows the UE **802** receiving an estimated TA. In an example, the first cell may be the deactivated cell **706** in FIG. 7. For example, **1206** may be performed by the TA validation component **198**.

(104) At **1208**, the UE performs a second measurement on the at least one DL-RS. For example, FIG. 8 at **818** shows the UE **802** performing second measurement(s) on a DL-RS. In another example, the second measurement may include a RSRP measurement performed at time t2 as illustrated in FIG. 7. For example, **1208** may be performed by the TA validation component **198**.

(105) At **1210**, the UE accepts or rejects the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. For example, FIG. 8 at **820** shows the UE **802** accepting or rejecting an estimated TA for communication with a cell based on at least one of first measurement(s) or second measurement(s). For example, **1210** may be performed by the TA validation component **198**.

(106) In one aspect, the estimated TA may be accepted or rejected further based on at least one of: a difference between the first measurement and the second measurement, or a RSRP threshold for the first cell associated with the second measurement. For example, FIG. 7 shows a threshold and a difference between a RSRP measurement performed at time t1 and a RSRP measurement performed at time t2.

(107) In one aspect, at **1202**, the UE may perform the first measurement on the at least one DL-RS prior to the first time instance, where the indication of the first measurement may be transmitted

based on the performed first measurement. For example, FIG. 8 at 810 shows the UE 802 performing first measurement(s) on a DL-RS associated with a cell. For example, 1202 may be performed by the TA validation component 198.

(108) In one aspect, the first cell may be a deactivated cell in the L1 mobility configured cell set or the L2 mobility configured cell set, where a second cell may serve as a SpCell prior to accepting or rejecting the estimated TA. For example, referring to FIG. 7, the first cell may be the deactivated cell 706 and the second cell may be the SpCell 704. In another example, the first cell and the second cell may be in the L1/L2 mobility configured cell set 508 illustrated in FIG. 5.

(109) In one aspect, the first measurement may be a DL timing difference measurement between the first cell and a reference cell, and the second measurement may be a RSRP measurement. For example, FIG. 6 shows a UE 602 measuring a DL timing difference with respect to a reference cell. In another example, FIG. 7 shows a UE 702 performing a RSRP measurement at time t2.

(110) In one aspect, at 1212, the UE may transmit a notification indicating that the estimated TA is rejected, where the notification may be transmitted if the estimated TA is rejected for the communication with the first cell. For example, FIG. 9 at 912 shows the UE 902 transmitting a TA invalid notification indicating that the estimated TA has been rejected for communication with the to-be-activated cell 906. For example, FIG. 10 at 1012 shows the UE 1002 transmitting a TA invalid notification indicating that the estimated TA has been rejected for communication with the to-be-activated cell 1006. For example, 1212 may be performed by the TA validation component 198.

(111) In one aspect, the notification may be transmitted via a MAC-CE or UCI. For example, FIG. 9 at 912 shows that the TA invalid notification may be transmitted via a MAC-CE or UCI. For example, FIG. 10 at 1012 shows that the TA invalid notification may be transmitted via a MAC-CE or UCI.

(112) In one aspect, the notification may be associated with an ACK or a NACK. For instance, FIG. 9 at 912 shows the TA invalid notification may be associated with an ACK/NACK message. For instance, FIG. 10 at 1012 shows that the TA invalid notification may be associated with an ACK/NACK message.

(113) In one aspect, at 1214, the UE may receive, subsequent to transmit the notification, a PDCCH order. For example, FIG. 9 at 914A shows the UE 902 receiving a PDCCH ordering a RACH procedure from the SpCell 904. In another example, FIG. 9 at 914B shows the UE 902 receiving a PDCCH ordering a RACH procedure from the to-be-activated cell 906. For example, 1214 may be performed by the TA validation component 198.

(114) In one aspect, at 1216, the UE may initiate a RACH procedure with the first cell based on the PDCCH order. For example, FIG. 9 at 916 shows the UE 902 initiating a RACH procedure with the to-be-activated cell 906. For example, 1216 may be performed by the TA validation component 198.

(115) In one aspect, the RACH procedure with the first cell may be initiated if the first cell is to serve as a SpCell. For example, FIG. 9 at 916 shows that the RACH procedure may be initiated if the to-be-activated cell 906 is to be a new SpCell.

(116) In one aspect, at 1218, the UE may perform, at a third time instance after the second time instance, a third measurement between the first cell and a reference cell, where the third measurement may be performed subsequent to transmitting the notification, and where the third measurement may be a DL timing difference measurement. For example, FIG. 10 at 1011 shows that the UE 1002 may perform measurement(s) after the TA validation at 1010, and the measurement(s) may include a DL timing difference measurement. The third measurement may be a DL timing difference measurement as illustrated in FIG. 7. For example, 1218 may be performed by the TA validation component 198.

(117) In one aspect, at 1220, the UE may transmit an indication of the third measurement via a MAC-CE or UCI. For example, FIG. 10 at 1012 shows the UE 1002 transmitting a TA invalid

notification via a MAC-CE or UCI, where the TA invalid notification may include UL sync assistance information. The UL sync assistance information may include the third measurement. In another example, the measurement transmitted by the UE **802** at **812** may include a third measurement. For example, **1220** may be performed by the TA validation component **198**.

(118) In one aspect, the third measurement may be included in the notification. For example, FIG. **10** at **1012** shows that the TA invalid notification may include UL sync assistance information. The UL sync assistance information may include the third measurement.

(119) In one aspect, at **1222**, the UE may receive, at a fourth time instance after the third time instance, a second estimated TA associated with the first cell. For example, FIG. **10** at **1016** shows that the UE **1002** may receive a new estimated TA. For example, **1222** may be performed by the TA validation component **198**.

(120) In one aspect, at **1224**, the UE may perform a fourth measurement on the at least one DL-RS. For example, FIG. **10** at **1018** shows that the UE **1002** may perform TA validation. The TA validation performed at **1018** may include performing a measurement as illustrated at **818** of FIG. **8**, where the measurement is a fourth measurement. For example, **1224** may be performed by the TA validation component **198**.

(121) In one aspect, at **1226**, the UE may accept or reject the second estimated TA for the communication with the first cell based on at least one of the third measurement or the fourth measurement. For example, FIG. **10** at **1018** shows that the UE **1002** may perform TA validation. The TA validation performed at **1018** may include accepting or rejecting an estimated TA as illustrated at **820** of FIG. **8**, where the estimated TA may be the second estimated TA. For example, **1226** may be performed by the TA validation component **198**.

(122) In one aspect, at **1228**, the UE may initiate a RACH procedure with the first cell based on a number of TAs being rejected exceeding a threshold number of rejections. For example, FIG. **10** at **1020** shows the UE **1002** determining if a maximum number of TA rejections has been reached and FIG. **10** at **1024** shows the UE **1002** initiating a RACH procedure if the maximum number of TA rejections has been reached. For example, **1228** may be performed by the TA validation component **198**.

(123) In one aspect, the first cell may serve as a SpCell or a SCell if the estimated TA is accepted. For example, the first cell may be the to-be-activated cell **906** or the to-be-activated cell **1006** and the to-be-activated cell **906** and the to-be-activated cell **1006** may serve as an SpCell or an SCell. In another example, the deactivated cell **706** in FIG. **7** may serve as an SpCell if the estimated TA is accepted.

(124) FIG. **13** is a flowchart **1300** of a method of wireless communication. The method may be performed by a network entity (e.g., the base station **102**, the CU **110**, the DU **130**, the RU **140**, the base station **310**, the base station **804**, the network entity **1502**, etc.). The method may be performed by the TA component **199**. The method may be associated with various technical advantages at the network entity, such as improved communications reliability via an accurate TA.

(125) At **1302**, the network entity receives, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. For example, FIG. **8** at **812** shows the base station **804** receiving first measurement(s). In an example, the first measurement(s) may be performed on a DL-RS associated with the deactivated cell **706** in FIG. **7**. In another example, the first measurement(s) may be a DL timing difference with respect to a reference cell at time t1 as illustrated in FIG. **6**. In yet another example, the first cell may be included in the L1/L2 mobility configured cell set **508** illustrated in FIG. **5**. For example, **1302** may be performed by the TA component **199**.

(126) At **1304**, the network entity estimates a TA associated with the first cell based on the first measurement on the at least one DL-RS. For example, FIG. **8** at **814** shows the base station **804** estimating a TA for a cell based on first measurement(s). For example, **1304** may be performed by

the TA component **199**.

(127) At **1306**, the network entity transmits, at a second time instance after the first time instance, the estimated TA associated with the first cell. For example, FIG. **8** at **816** shows the base station **804** transmitting an estimated TA. In an example, the first cell may be the deactivated cell **706** in FIG. **7**. For example, **1306** may be performed by the TA component **199**.

(128) FIG. **14** is a flowchart **1400** of a method of wireless communication. The method may be performed by a network entity (e.g., the base station **102**, the CU **110**, the DU **130**, the RU **140**, the base station **310**, the base station **804**, the network entity **1502**, etc.). The method (including the various aspects described below) may be performed by the TA component **199**. The method may be associated with various technical advantages at the network entity, such as improved communications reliability via an accurate TA.

(129) At **1402**, the network entity receives, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. For example, FIG. **8** at **812** shows the base station **804** receiving first measurement(s). In an example, the first measurement(s) may be performed on a DL-RS associated with the deactivated cell **706** in FIG. **7**. In another example, the first measurement(s) may be a DL timing difference with respect to a reference cell at time t_1 as illustrated in FIG. **6**. In yet another example, the first cell may be included in the L1/L2 mobility configured cell set **508** illustrated in FIG. **5**. For example, **1402** may be performed by the TA component **199**.

(130) At **1404**, the network entity estimates a TA associated with the first cell based on the first measurement on the at least one DL-RS. For example, FIG. **8** at **814** shows the base station **804** estimating a TA for a cell based on first measurement(s). For example, **1404** may be performed by the TA component **199**.

(131) At **1406**, the network entity transmits, at a second time instance after the first time instance, the estimated TA associated with the first cell. For example, FIG. **8** at **816** shows the base station **804** transmitting an estimated TA. In an example, the first cell may be the deactivated cell **706** in FIG. **7**. For example, **1406** may be performed by the TA component **199**.

(132) In one aspect, the first cell may be a deactivated cell in the L1 mobility configured cell set or the L2 mobility configured cell set, and a second cell may serve as a SpCell. For example, referring to FIG. **7**, the first cell may be the deactivated cell **706** and the second cell may be the SpCell **704**. In another example, the first cell and the second cell may be in the L1/L2 mobility configured cell set **508** illustrated in FIG. **5**.

(133) In one aspect, the first measurement may be a DL timing difference measurement between the first cell and a reference cell. For example, FIG. **6** shows a DL timing difference measurement with respect to a reference cell.

(134) In one aspect, at **1408**, the network entity may receive a notification indicating that the estimated TA has been rejected for communication with the first cell. For example, FIG. **9** at **912** shows the SpCell **904** receiving a TA invalid notification indicating that the estimated TA has been rejected for communication with the to-be-activated cell **906**. For example, FIG. **10** at **1012** shows the SpCell **1004** receiving a TA invalid notification indicating that the estimated TA has been rejected for communication with the to-be-activated cell **1006**. For example, **1408** may be performed by the TA component **199**.

(135) In one aspect, the notification may be received via a MAC-CE or UCI. For example, FIG. **9** at **912** shows that the TA invalid notification may be received via a MAC-CE or UCI. For example, FIG. **10** at **1012** shows that the TA invalid notification may be received via a MAC-CE or UCI.

(136) In one aspect, the notification may be associated with an ACK or a NACK. For instance, FIG. **9** at **912** shows the TA invalid notification may be associated with an ACK/NACK message. For instance, FIG. **10** at **1012** shows the TA invalid notification may be associated with an ACK/NACK message.

(137) In one aspect, at **1410**, the network entity may transmit a PDCCH order for initiating a RACH procedure with the first cell based on receiving the notification. For example, FIG. **9** at **914A** shows the SpCell **904** transmitting a PDCCH order for initiating a RACH procedure with the to-be-activated cell **906**. For example, **1410** may be performed by the TA component **199**.

(138) In one aspect, at **1412**, the network entity may receive an indication of a third measurement at a third time instance after the second time instance, where the indication of the third measurement may be received via a MAC-CE or UCI, and where the third measurement may be a DL timing difference measurement between the first cell and a reference cell. For example, FIG. **10** at **1012** shows the SpCell **1004** receiving a TA invalid notification including UL sync assistance information. The UL sync assistance information may include a third measurement, where the third measurement is a DL timing difference. FIG. **10** at **1012** also shows the third measurement may be received via a MAC-CE or UCI. In yet another example, the SpCell **1004** may receive the third measurement separately. For example, **1412** may be performed by the TA component **199**.

(139) In one aspect, the third measurement may be included in the notification. For example, FIG. **10** at **1012** shows that the TA invalid notification may include UL sync assistance information that may include the measurement.

(140) In one aspect, at **1414**, the network entity may estimate a second TA associated with the first cell based on the third measurement. For example, FIG. **10** at **1014** shows the SpCell **1004** estimating a new TA based on the UL sync assistance information. For example, **1414** may be performed by the TA component **199**.

(141) In one aspect, at **1416**, the network entity may transmit, at a fourth time instance after the second time instance, the second estimated TA associated with the first cell. For example, FIG. **10** at **1016** shows the SpCell **1004** transmitting a new estimated TA for the to-be-activated cell **1006**. For example, **1416** may be performed by the TA component **199**.

(142) In one aspect, the first cell may serve as a new SpCell or a SCell if the estimated TA is accepted. For example, the first cell may be the to-be-activated cell **906** or the to-be-activated cell **1006** and the to-be-activated cell **906** and the to-be-activated cell **1006** may serve as an SpCell or an SCell. In another example, the deactivated cell **706** in FIG. **7** may serve as an SpCell if the estimated TA is accepted.

(143) FIG. **15** is a diagram **1500** illustrating an example of a hardware implementation for an apparatus **1504**. The apparatus **1504** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **1504** may include a cellular baseband processor **1524** (also referred to as a modem) coupled to one or more transceivers **1522** (e.g., cellular RF transceiver). The cellular baseband processor **1524** may include on-chip memory **1524'**. In some aspects, the apparatus **1504** may further include one or more subscriber identity modules (SIM) cards **1520** and an application processor **1506** coupled to a secure digital (SD) card **1508** and a screen **1510**. The application processor **1506** may include on-chip memory **1506'**. In some aspects, the apparatus **1504** may further include a Bluetooth module **1512**, a WLAN module **1514**, an SPS module **1516** (e.g., GNSS module), one or more sensor modules **1518** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial measurement unit (IMU), gyroscope, and/or accelerometer(s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other technologies used for positioning), additional memory modules **1526**, a power supply **1530**, and/or a camera **1532**. The Bluetooth module **1512**, the WLAN module **1514**, and the SPS module **1516** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **1512**, the WLAN module **1514**, and the SPS module **1516** may include their own dedicated antennas and/or utilize the antennas **1580** for communication. The cellular baseband processor **1524** communicates through the transceiver(s) **1522** via one or more antennas **1580** with the UE **104** and/or with an RU associated with a network entity **1502**. The cellular baseband processor **1524** and the application processor **1506** may each include a computer-readable medium/memory

1524', **1506'**, respectively. The additional memory modules **1526** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory **1524'**, **1506'**, **1526** may be non-transitory. The cellular baseband processor **1524** and the application processor **1506** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor **1524**/application processor **1506**, causes the cellular baseband processor **1524**/application processor **1506** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor **1524**/application processor **1506** when executing software. The cellular baseband processor **1524**/application processor **1506** may be a component of the UE **350** and may include the memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **1504** may be a processor chip (modem and/or application) and include just the cellular baseband processor **1524** and/or the application processor **1506**, and in another configuration, the apparatus **1504** may be the entire UE (e.g., see **350** of FIG. 3) and include the additional modules of the apparatus **1504**.

(144) As discussed supra, the TA validation component **198** is configured to transmit, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. The TA validation component **198** is also configured to receive, at a second time instance after the first time instance, an estimated TA associated with the first cell. The TA validation component **198** is also configured to perform a second measurement on the at least one DL-RS. The TA validation component **198** is also configured to accept or reject the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. The TA validation component **198** may be within the cellular baseband processor **1524**, the application processor **1506**, or both the cellular baseband processor **1524** and the application processor **1506**. The TA validation component **198** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. As shown, the apparatus **1504** may include a variety of components configured for various functions. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for transmitting, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for receiving, at a second time instance after the first time instance, an estimated TA associated with the first cell. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for performing a second measurement on the at least one DL-RS. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for accepting or rejecting the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for performing the first measurement on the at least one DL-RS prior to the first time instance, where the indication of the first measurement is transmitted based on the performed first measurement. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for transmitting a notification indicating that the estimated TA is rejected, where the notification is transmitted if the estimated TA is rejected for the communication with the first cell. In one configuration, the apparatus **1504**, and in particular the cellular baseband

processor **1524** and/or the application processor **1506**, includes means for receiving, subsequent to transmitting the notification, a PDCCH order. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for initiating a RACH procedure with the first cell based on the PDCCH order. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for performing, at a third time instance after the second time instance, a third measurement between the first cell and a reference cell, where the third measurement is performed subsequent to transmitting the notification, and where the third measurement is a DL timing difference measurement. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for transmitting an indication of the third measurement via a MAC-CE or UCI. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for receiving, at a fourth time instance after the third time instance, a second estimated TA associated with the first cell. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for performing a fourth measurement on the at least one DL-RS. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for accepting or rejecting the second estimated TA for the communication with the first cell based on at least one of the third measurement or the fourth measurement. In one configuration, the apparatus **1504**, and in particular the cellular baseband processor **1524** and/or the application processor **1506**, includes means for initiating a RACH procedure with the first cell based on a number of TAs being rejected exceeding a threshold number of rejections. The means may be the TA validation component **198** of the apparatus **1504** configured to perform the functions recited by the means. As described supra, the apparatus **1504** may include the TX processor **368**, the RX processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX processor **368**, the RX processor **356**, and/or the controller/processor **359** configured to perform the functions recited by the means.

(145) FIG. **16** is a diagram **1600** illustrating an example of a hardware implementation for a network entity **1602**. The network entity **1602** may be a BS, a component of a BS, or may implement BS functionality. The network entity **1602** may include at least one of a CU **1610**, a DU **1630**, or an RU **1640**. For example, depending on the layer functionality handled by the TA component **199**, the network entity **1602** may include the CU **1610**; both the CU **1610** and the DU **1630**; each of the CU **1610**, the DU **1630**, and the RU **1640**; the DU **1630**; both the DU **1630** and the RU **1640**; or the RU **1640**. The CU **1610** may include a CU processor **1612**. The CU processor **1612** may include on-chip memory **1612'**. In some aspects, the CU **1610** may further include additional memory modules **1614** and a communications interface **1618**. The CU **1610** communicates with the DU **1630** through a midhaul link, such as an F1 interface. The DU **1630** may include a DU processor **1632**. The DU processor **1632** may include on-chip memory **1632'**. In some aspects, the DU **1630** may further include additional memory modules **1634** and a communications interface **1638**. The DU **1630** communicates with the RU **1640** through a fronthaul link. The RU **1640** may include an RU processor **1642**. The RU processor **1642** may include on-chip memory **1642'**. In some aspects, the RU **1640** may further include additional memory modules **1644**, one or more transceivers **1646**, antennas **1680**, and a communications interface **1648**. The RU **1640** communicates with the UE **104**. The on-chip memory **1612'**, **1632'**, **1642'** and the additional memory modules **1614**, **1634**, **1644** may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. Each of the processors **1612**, **1632**, **1642** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various functions described

supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

(146) As discussed supra, the TA component **199** is configured to receive, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. The TA component **199** is configured to estimate a TA associated with the first cell based on the first measurement on the at least one DL-RS. The TA component **199** is configured to transmit, at a second time instance after the first time instance, the estimated TA associated with the first cell. The TA component **199** may be within one or more processors of one or more of the CU **1610**, DU **1630**, and the RU **1640**. The TA component **199** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof. The network entity **1602** may include a variety of components configured for various functions. In one configuration, the network entity **1602** includes means for receiving, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. In one configuration, the network entity **1602** includes means for estimating a TA associated with the first cell based on the first measurement on the at least one DL-RS. In one configuration, the network entity **1602** includes means for transmitting, at a second time instance after the first time instance, the estimated TA associated with the first cell. In one configuration, the network entity **1602** includes means for receiving a notification indicating that the estimated TA has been rejected for communication with the first cell. In one configuration, the network entity **1602** includes means for transmitting a PDCCH order for initiating a RACH procedure with the first cell based on receiving the notification. In one configuration, the network entity **1602** includes means for receiving an indication of a third measurement at a third time instance after the second time instance, where the indication of the third measurement is received via a MAC-CE or UCI, and where the third measurement is a DL timing difference measurement between the first cell and a reference cell. In one configuration, the network entity **1602** includes means for estimating a second TA associated with the first cell based on the third measurement. In one configuration, the network entity **1602** includes means for transmitting, at a fourth time instance after the second time instance, the second estimated TA associated with the first cell. The means may be the TA component **199** of the network entity **1602** configured to perform the functions recited by the means. As described supra, the network entity **1602** may include the TX processor **316**, the RX processor **370**, and the controller/processor **375**. As such, in one configuration, the means may be the TX processor **316**, the RX processor **370**, and/or the controller/processor **375** configured to perform the functions recited by the means.

(147) A UE may apply a TA for UL transmissions with a cell to account for propagation delays between the UE and the cell. When the UE receives signaling from a network to connect to a to-be-activated cell in a L1/L2 mobility configured cell set, the network may provide an estimated TA to the UE, where the estimated TA may be based on a measurement report transmitted by the UE to the network. However, the estimated TA may be inaccurate for various reasons. In one example, channel strength may have changed between a time at which the UE last transmitted a measurement report and a time at which the estimated TA is obtained, which may lead to the estimated TA being inaccurate. In another example, the UE may not transmit measurement reports frequently enough to enable the network to accurately generate the estimated TA. In yet another example, the estimated TA may be inaccurate due to UE mobility, that is, a time at which the UE applies the estimated TA may be too far away from a time at which the network generates the estimated TA. Applying an inaccurate TA for UL transmissions may impact communications reliability.

(148) Various techniques for UE-side TA validation are described herein that may utilize RSRP measurements to represent channel strength. In an example, a UE transmits, at a first time instance, an indication of a first measurement on at least one DL-RS associated with a first cell, where the first cell is included in a L1 mobility configured cell set or a L2 mobility configured cell set. The UE receives, at a second time instance after the first time instance, an estimated TA associated with the first cell. The UE performs a second measurement on the at least one DL-RS. The UE accepts or rejects the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement. Via the aforementioned TA validation procedure, the UE may be able to reject inaccurate TAs. As such, the TA validation procedure may lead to increased communications reliability.

(149) It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not limited to the specific order or hierarchy presented.

(150) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be accorded the full scope consistent with the language claims. Reference to an element in the singular does not mean “one and only one” unless specifically so stated, but rather “one or more.” Terms such as “if,” “when,” and “while” do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., “when,” do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. If a first apparatus receives data from or transmits data to a second apparatus, the data may be received/transmitted directly between the first and second apparatuses, or indirectly between the first and second apparatuses through a set of apparatuses. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. Moreover, nothing disclosed herein is dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.”

(151) As used herein, the phrase “based on” shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase “based on A” (where “A” may be information, a condition, a factor, or the like) shall be construed

as “based at least on A” unless specifically recited differently.

(152) The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation.

(153) Aspect 1 is a method of wireless communication at a user equipment (UE), including: transmitting, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, where the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; receiving, at a second time instance after the first time instance, an estimated timing advance (TA) associated with the first cell; performing a second measurement on the at least one DL-RS; and accepting or rejecting the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement.

(154) Aspect 2 is the method of aspect 1, where the estimated TA is accepted or rejected further based on at least one of: a difference between the first measurement and the second measurement, or a reference signal received power (RSRP) threshold for the first cell associated with the second measurement.

(155) Aspect 3 is the method of any of aspects 1-2, further including: performing the first measurement on the at least one DL-RS prior to the first time instance, where the indication of the first measurement is transmitted based on the performed first measurement.

(156) Aspect 4 is the method of any of aspects 1-3, where the first cell is a deactivated cell in the L1 mobility configured cell set or the L2 mobility configured cell set, where a second cell is serving as a special cell (SpCell) prior to accepting or rejecting the estimated TA.

(157) Aspect 5 is the method of any of aspects 1-4, where the first measurement is a DL timing difference measurement between the first cell and a reference cell, and where the second measurement is a reference signal received power (RSRP) measurement.

(158) Aspect 6 is the method of any of aspects 1-5, further including: transmitting a notification indicating that the estimated TA is rejected, where the notification is transmitted if the estimated TA is rejected for the communication with the first cell.

(159) Aspect 7 is the method of aspect 6, where the notification is transmitted via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI).

(160) Aspect 8 is the method of aspect 6, where the notification is associated with an acknowledgement (ACK) or a negative acknowledgment (NACK).

(161) Aspect 9 is the method of aspects 6-7 or aspects 6 and 8, further including: receiving, subsequent to transmitting the notification, a physical downlink control channel (PDCCH) order; and initiating a random access channel (RACH) procedure with the first cell based on the PDCCH order.

(162) Aspect 10 is the method of aspect 9, where the RACH procedure with the first cell is initiated if the first cell is to serve as a special cell (SpCell).

(163) Aspect 11 is the method of aspect 6, further including: performing, at a third time instance after the second time instance, a third measurement between the first cell and a reference cell, where the third measurement is performed subsequent to transmitting the notification, and where the third measurement is a DL timing difference measurement.

(164) Aspect 12 is the method of aspect 11, further including: transmitting an indication of the third measurement via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI).

(165) Aspect 13 is the method of aspect 11, where the third measurement is included in the notification.

(166) Aspect 14 is the method of aspects 11-12 or aspects 11 and 13, further including: receiving, at a fourth time instance after the third time instance, a second estimated TA associated with the first cell; performing a fourth measurement on the at least one DL-RS; and accepting or rejecting the second estimated TA for the communication with the first cell based on at least one of the third

measurement or the fourth measurement.

(167) Aspect 15 is the method of aspect 14, further including: initiating a random access channel (RACH) procedure with the first cell based on a number of TAs being rejected exceeding a threshold number of rejections.

(168) Aspect 16 is the method of any of aspects 1-15 where the first cell serves as a new special cell (SpCell) or a secondary cell (SCell) if the estimated TA is accepted.

(169) Aspect 17 is an apparatus for wireless communication at a user equipment (UE) including a memory and at least one processor coupled to the memory and based at least in part on information stored in the memory, the at least one processor is configured to perform a method in accordance with any of aspects 1-16.

(170) Aspect 18 is an apparatus for wireless communications, including means for performing a method in accordance with any of aspects 1-16.

(171) Aspect 19 is the apparatus of aspect 17 or 18 further including at least one of a transceiver or an antenna coupled to the at least one processor, where at least one of the transceiver or the antenna is configured to transmit the indication of the first measurement and receive the estimated TA associated with the first cell.

(172) Aspect 20 is a non-transitory computer-readable medium including instructions that, when executed by an apparatus, cause the apparatus to perform a method in accordance with any of aspects 1-16.

(173) Aspect 21 is a method of wireless communication at a network entity, including: receiving, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, where the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; estimating a timing advance (TA) associated with the first cell based on the first measurement on the at least one DL-RS; and transmitting, at a second time instance after the first time instance, the estimated TA associated with the first cell.

(174) Aspect 22 is the method of aspect 21, where the first cell is a deactivated cell in the L1 mobility configured cell set or the L2 mobility configured cell set, where a second cell is serving as a special cell (SpCell).

(175) Aspect 23 is the method of any of aspects 21-22, where the first measurement is a DL timing difference measurement between the first cell and a reference cell.

(176) Aspect 24 is the method of any of aspects 21-23, further including: receiving a notification indicating that the estimated TA has been rejected for communication with the first cell.

(177) Aspect 25 is the method of aspect 24, where the notification is received via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI).

(178) Aspect 26 is the method of aspect 24, where the notification is associated with an acknowledgement (ACK) or a negative acknowledgment (NACK).

(179) Aspect 27 is the method of aspects 24-25 or aspects 24 and 26, further including: transmitting a physical downlink control channel (PDCCH) order for initiating a random access channel (RACH) procedure with the first cell based on receiving the notification.

(180) Aspect 28 is the method of aspect 24, further including: receiving an indication of a third measurement at a third time instance after the second time instance, where the indication of the third measurement is received via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI), and where the third measurement is a DL timing difference measurement between the first cell and a reference cell.

(181) Aspect 29 is the method of aspect 28, where the third measurement is included in the notification.

(182) Aspect 30 is the method of any of aspects 28-29, further including: estimating a second TA associated with the first cell based on the third measurement; and transmitting, at a fourth time instance after the second time instance, the second estimated TA associated with the first cell.

- (183) Aspect 31 is the method of any of aspects 21-30 where the first cell serves as a new special cell (SpCell) or a secondary cell (SCell) if the estimated TA is accepted.
- (184) Aspect 32 is an apparatus for wireless communication at a user equipment (UE) including a memory and at least one processor coupled to the memory and based at least in part on information stored in the memory, the at least one processor is configured to perform a method in accordance with any of aspects 21-31.
- (185) Aspect 33 is an apparatus for wireless communications, including means for performing a method in accordance with any of aspects 21-31.
- (186) Aspect 34 is the apparatus of aspect 32 or 33 further including at least one of a transceiver or an antenna coupled to the at least one processor, where at least one of the transceiver or the antenna is configured to receive the indication of the first measurement and transmit the estimated TA.
- (187) Aspect 35 is a non-transitory computer-readable medium including instructions that, when executed by an apparatus, cause the apparatus to perform a method in accordance with any of aspects 21-31.

Claims

1. An apparatus for wireless communication at a user equipment (UE), comprising: memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: transmit, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, wherein the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; receive, at a second time instance after the first time instance, an estimated timing advance (TA) associated with the first cell; perform a second measurement on the at least one DL-RS; and accept or reject the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement.
2. The apparatus of claim 1, wherein the estimated TA is accepted or rejected further based on at least one of: a difference between the first measurement and the second measurement, or a reference signal received power (RSRP) threshold for the first cell associated with the second measurement.
3. The apparatus of claim 1, wherein the at least one processor is further configured to: perform the first measurement on the at least one DL-RS prior to the first time instance, wherein the indication of the first measurement is transmitted based on the performed first measurement.
4. The apparatus of claim 1, wherein the first cell is a deactivated cell in the L1 mobility configured cell set or the L2 mobility configured cell set, wherein a second cell is serving as a special cell (SpCell) prior to accept or reject the estimated TA.
5. The apparatus of claim 1, wherein the first measurement is a DL timing difference measurement between the first cell and a reference cell, and wherein the second measurement is a reference signal received power (RSRP) measurement.
6. The apparatus of claim 1, wherein the at least one processor is further configured to: transmit a notification indicating that the estimated TA is rejected, wherein the notification is transmitted if the estimated TA is rejected for the communication with the first cell.
7. The apparatus of claim 6, wherein the notification is transmitted via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI).
8. The apparatus of claim 6, wherein the notification is associated with an acknowledgement (ACK) or a negative acknowledgment (NACK).
9. The apparatus of claim 6, wherein the at least one processor is further configured to: receive, subsequent to transmit the notification, a physical downlink control channel (PDCCH) order; and initiate a random access channel (RACH) procedure with the first cell based on the PDCCH order.
10. The apparatus of claim 9, wherein the RACH procedure with the first cell is initiated if the first

cell is to serve as a special cell (SpCell).

11. The apparatus of claim 6, wherein the at least one processor is further configured to: perform, at a third time instance after the second time instance, a third measurement between the first cell and a reference cell, wherein the third measurement is performed subsequent to transmit the notification, and wherein the third measurement is a DL timing difference measurement.

12. The apparatus of claim 11, wherein the at least one processor is further configured to: transmit an indication of the third measurement via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI).

13. The apparatus of claim 11, wherein the third measurement is included in the notification.

14. The apparatus of claim 11, wherein the at least one processor is further configured to: receive, at a fourth time instance after the third time instance, a second estimated TA associated with the first cell; perform a fourth measurement on the at least one DL-RS; and accept or reject the second estimated TA for the communication with the first cell based on at least one of the third measurement or the fourth measurement.

15. The apparatus of claim 14, wherein the at least one processor is further configured to: initiate a random access channel (RACH) procedure with the first cell based on a number of TAs being rejected exceeding a threshold number of rejections.

16. The apparatus of claim 1, wherein the first cell serves as a new special cell (SpCell) or a secondary cell (SCell) if the estimated TA is accepted.

17. The apparatus of claim 1, further comprising at least one of a transceiver or an antenna coupled to the at least one processor, wherein at least one of the transceiver or the antenna is configured to: transmit the indication of the first measurement and receive the estimated TA.

18. A method of wireless communication at a user equipment (UE), comprising: transmitting, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, wherein the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; receiving, at a second time instance after the first time instance, an estimated timing advance (TA) associated with the first cell; performing a second measurement on the at least one DL-RS; and accepting or rejecting the estimated TA for communication with the first cell based on at least one of the first measurement or the second measurement.

19. An apparatus for wireless communication at a network entity, comprising: memory; and at least one processor coupled to the memory and, based at least in part on information stored in the memory, the at least one processor is configured to: receive, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, wherein the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; estimate a timing advance (TA) associated with the first cell based on the first measurement on the at least one DL-RS; and transmit, at a second time instance after the first time instance, the estimated TA associated with the first cell.

20. The apparatus of claim 19, wherein the first cell is a deactivated cell in the L1 mobility configured cell set or the L2 mobility configured cell set, wherein a second cell is serving as a special cell (SpCell).

21. The apparatus of claim 19, wherein the first measurement is a DL timing difference measurement between the first cell and a reference cell.

22. The apparatus of claim 19, wherein the at least one processor is further configured to: receive a notification indicating that the estimated TA has been rejected for communication with the first cell.

23. The apparatus of claim 22, wherein the notification is received via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI).

24. The apparatus of claim 22, wherein the notification is associated with an acknowledgement (ACK) or a negative acknowledgment (NACK).

25. The apparatus of claim 22, wherein the at least one processor is further configured to: transmit

a physical downlink control channel (PDCCH) order for initiating a random access channel (RACH) procedure with the first cell based on receive the notification.

26. The apparatus of claim 22, wherein the at least one processor is further configured to: receive an indication of a third measurement at a third time instance after the second time instance, wherein the indication of the third measurement is received via a medium access control (MAC) control element (MAC-CE) or uplink control information (UCI), and wherein the third measurement is a DL timing difference measurement between the first cell and a reference cell.

27. The apparatus of claim 26, wherein the third measurement is included in the notification.

28. The apparatus of claim 26, wherein the at least one processor is further configured to: estimate a second TA associated with the first cell based on the third measurement; and transmit, at a fourth time instance after the second time instance, the second estimated TA associated with the first cell.

29. The apparatus of claim 19, further comprising at least one of a transceiver or an antenna coupled to the at least one processor, wherein at least one of the transceiver or the antenna is configured to: receive the indication of the first measurement and transmit the estimated TA.

30. A method of wireless communication at a network entity, comprising: receiving, at a first time instance, an indication of a first measurement on at least one downlink reference signal (DL-RS) associated with a first cell, wherein the first cell is included in a layer 1 (L1) mobility configured cell set or a layer 2 (L2) mobility configured cell set; estimating a timing advance (TA) associated with the first cell based on the first measurement on the at least one DL-RS; and transmitting, at a second time instance after the first time instance, the estimated TA associated with the first cell.
